Derek Phung

B.S. Aerospace Engineering Student · University of California, Irvine

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Education

University of California, Irvine *Irvine, CA*

B.S. in Aerospace Engineering *2023.9 - Winter 2027 (expected)*

- eXploration Logistics and Control (XLC) Group (Advisor: Prof. Yuri Shimane)
- GPA: 3.353 / 4.000. Graduate-level coursework: **Space Systems Optimization (A+)**

Publications

(C1) **Phung, D.**, and Shimane, Y. "High-Fidelity Baseline Design and Station Keeping Analyses for Earth–Moon Vertical Orbits," accepted for oral presentation at the 2027 AAS Space Flight Mechanics Meeting; arXiv:2610.10803, 2026.

(J1) Shimane, Y., and **Phung, D.** "HighFidelityEphemerisModel.jl," software manuscript in preparation for the *Journal of Open Source Software*.

Experiences

Undergraduate Researcher / UROP Fellow *Remote / Irvine, CA*

University of California, Irvine, XLC Group (Advisor: Prof. Yuri Shimane) *2026.1 - Present*

Geometry-Preserving High-Fidelity Baseline Generation and Model Predictive Station Keeping *2026.1 - Present*

- Generated eight 5.5-year Earth–Moon L1/L2 vertical-orbit baselines, including six ballistic solutions, and 200 six-month family baselines by mapping CR3BP trajectories into high-fidelity ephemeris dynamics and refining them with sequential convex programming while preserving orbit geometry.
- Developed model predictive station keeping and evaluated robustness through five-year, 100-seed Monte Carlo campaigns across all eight long-duration cases, along with one-year controller/noise sensitivity studies.
- Characterized long-duration orbit behavior through finite-time Lyapunov exponent (FTLE) / stretching analysis, epoch and geometry-constraint studies, and SPICE SPK/BSP validation via direct and propagated-state comparisons.

High-Fidelity Astrodynamics Software Development and Validation *2026.5 - Present*

- Implemented Gauss variational equation (GVE) propagation, a native Ephemerides backend, and SPK trajectory utilities/workflows in **HighFidelityEphemerisModel.jl**.
- Validated the GVE and Ephemerides implementations internally against the SPICE backend and performed external HFEM validation against GMAT, Orekit, NBodyPropagator.jl, Nyx/ANISE, and Basilisk using matched physical models and data.
- Built numerical tests and cross-formulation comparisons for Cartesian, modified-equinoctial, Keplerian, and ordinary-equinoctial propagation across shared domains and representative cislunar and lunar-orbit cases.

Ordinary Equinoctial Orbital-Element Support *2026.7 - 2026.8*

- Implemented ordinary equinoctial element support in **AstrodynamicsCore.jl**, adding Keplerian, modified-equinoctial, and Cartesian conversion paths, domain-aware validation, round-trip state checks, and API/tutorial documentation.
- Validated conversions across circular, inclined, eccentric, near-equatorial, and near-retrograde cases, including expected failures for singular and non-elliptic states.

Mission Assurance - NASA L'SPACE Mission Concept Academy *Remote*

NASA L'SPACE Program *2026.1 - 2026.5*

- Performed mission assurance for a multidisciplinary space mission concept, developing risk matrices, mitigation strategies, and hazard controls incorporated into MCR, SRR, MDR, and PDR deliverables.

Fuselage Engineering Team Member *Irvine, CA*

Flying Leatherneck Aviation Museum *2025.9 - 2026.4*

- Supported reconstruction of a 1918 Curtiss JN-4 "Jenny" aircraft through fabrication, alignment, and SolidWorks modeling of structural and mechanical components from historical drawings.

Pneumatic Autonomous Rover - Controls Lead *Irvine, CA*

University of California, Irvine *2025.1 - 2025.3*

- Designed an Arduino-based PID controller for autonomous checkpoint navigation, implemented low-pass filtering for magnetometer noise, and tuned closed-loop steering through iterative testing.

Autonomous Rover - Programmer and Researcher *Irvine, CA*

University of California, Irvine *2024.1 - 2024.3*

- Integrated LiDAR and PixyCam sensing for path alignment and object retrieval, designed the chassis in SolidWorks, and achieved a top-10 completion time in the course competition.

Remote-Controlled Rover - Team Lead and Mechanical Designer *Irvine, CA*

University of California, Irvine *2023.9 - 2023.12*

- Led a six-member team in rover design and fabrication, including chassis layout and Ackermann steering geometry.

Awards, Programs, & Activities

2023 Fall **Dean's Honor List**, UC Irvine

2024 Spring **Dean's Honor List**, UC Irvine

2025 Fall **Dean's Honor List**, UC Irvine

Miscellaneous

RESEARCH INTERESTS: Astrodynamics; cislunar dynamics; spacecraft guidance, navigation, and control; trajectory optimization; autonomous spacecraft systems; high-fidelity modeling and simulation.

SKILLS: **Programming:** Julia, MATLAB, Python, C++/Arduino, R.
Astrodynamics: SPICE, JPL ephemerides, CR3BP dynamics, N-body propagation, lunar spherical-harmonic gravity, orbital-element propagation.
Optimization & Control: sequential convex programming / successive convexification, model predictive control, trajectory optimization, Monte Carlo analysis.
Software Development: Git, GitHub, scientific software development, numerical validation and testing, LaTeX/Overleaf.
Engineering: SolidWorks, Siemens NX, Onshape.

LANGUAGES: English (native), Japanese (conversational).